

## VIVA READY NOTE

# QuizMaster / Online Quiz Management System

Compact Bangla note for viva. Style: day-to-day explanation + a little English technical term so you can explain confidently and naturally.

Express.js

EJS

MongoDB + Mongoose

Passport Auth

Role Based Access

Quiz + Coding Module

## 1 minute introduction

“আমার project টা একটা **role-based online quiz management system**। এখানে **admin** teacher manage করে, **teacher** quiz create/publish করে, আর **student** enroll করে exam দেয়, result দেখে, leaderboard আর progress track করে। আমি শুধু MCQ না, **true/false**, **short answer**, আর **coding test / coding submission** flow-ও রেখেছি। পুরো system টা Express + EJS + MongoDB দিয়ে build করা হয়েছে।”

## Quick Identity

|                     |                                              |
|---------------------|----------------------------------------------|
| <b>Project type</b> | Web application                              |
| <b>Architecture</b> | MVC style                                    |
| <b>Frontend</b>     | EJS + CSS + Bootstrap/Tailwind utility usage |
| <b>Backend</b>      | Express.js                                   |
| <b>Database</b>     | MongoDB                                      |
| <b>Auth</b>         | Passport local + Google                      |

## Main modules I built

### Admin Panel

- First admin setup
- Create other admin
- Approve / reject teacher
- Grant teacher by email
- Block / unblock student or teacher

Control Layer

### Teacher Panel

- Create, edit, publish quiz
- Add multiple question types
- Review pending answers
- View analytics + leaderboard
- Manage coding problems/submissions

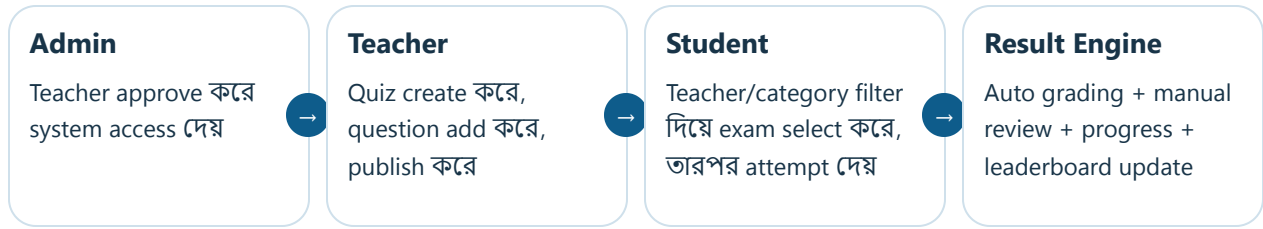
Content Creation

### Student Panel

- Browse exam and enroll
- Attempt exam
- See result/history/review
- Progress + global leaderboard
- Teacher-wise and category-wise filter

Assessment Experience

## System flow visual



## Simple explanation I can say in viva

“Real life-এ অনেক teacher থাকলে student confuse হয়ে যায়। তাই আমি শুধু exam list show করি নাই, বরং **teacher-wise filter** আর **category-wise grouping** add করেছি। Student আগে teacher select করতে পারে, তারপর ওই teacher-এর published exam গুলো দেখতে পারে। এতে UX clear হয় এবং intended teacher-এর exam খুঁজে পাওয়া easy হয়।”

## ARCHITECTURE

## Project structure I can explain

## Folder responsibility

| Folder       | What it does                                  |
|--------------|-----------------------------------------------|
| routes/      | URL define করে, controller call করে           |
| controllers/ | Business logic handle করে                     |
| models/      | Database schema ও relation define করে         |
| views/       | EJS templates, UI rendering                   |
| middleware/  | Auth, tab session, error handling             |
| utils/       | Progress/points calculation type helper logic |

## Why MVC style?

- Code clean থাকে
- Logic আর UI আলাদা থাকে
- Maintenance easy হয়
- New feature add করা সহজ
- Controller reuse করা যায়

Viva line: “আমি MVC-like separation follow করেছি যাতে route, business logic, database schema আর view একসাথে জট না পাকায়।”

## Database entities and relations

| Model      | Main fields                                                        | Purpose                                    |
|------------|--------------------------------------------------------------------|--------------------------------------------|
| User       | name, email, role, teacherStatus, accountStatus                    | Admin/Teacher/Student identity manage করে  |
| Quiz       | title, category, examType, difficulty, duration, createdBy, status | Teacher-created exam container             |
| Question   | quiz, type, options, correctAnswer, marks, testCases               | Different question type store করে          |
| Enrollment | student, quiz, status, attempts, bestScore                         | Student কোন exam-এ enrolled সেটা track করে |
| Attempt    | student, quiz, answers, score, percentage, status                  | Exam submission record                     |
| Progress   | averageScore, totalPoints, streak, quizzesByCategory               | Student progress analytics                 |

|                                 |                                     |                                         |
|---------------------------------|-------------------------------------|-----------------------------------------|
| Leaderboard / GlobalLeaderboard | bestPercentage, rank, totalPoints   | Quiz-wise and platform-wise ranking     |
| Problem / Submission            | coding problem, code, review, marks | Coding practice and teacher review flow |

## Key features in plain Bangla

### Authentication

Local login + Google login, password hash with bcrypt, session stored in MongoDB.

### Teacher approval

সব teacher auto-access পায় না। Admin approve করলে তবেই teacher dashboard use করতে পারে।

### Blocked user control

Admin account block করলে user next request থেকেই access হারায়।

### Mixed question support

MCQ, true/false, short-answer, coding type আলাদা validation দিয়ে handle করা হয়েছে।

### Teacher review flow

Short answer / coding manual review লাগলে teacher marks আর comment add করে।

### Smart browsing

Student এখন teacher-wise, category-wise, difficulty-wise, exam-type-wise filter করতে পারে।

## Teacher to student data flow

### Create

Teacher quiz create করে with category, examType, difficulty



### Publish

Status draft থেকে published হয়



### Enroll & Attempt

Student enroll করে exam submit দেয়



### Update records

Attempt, Result, Progress, Leaderboard সব update হয়

## CORE LOGIC

## How exam evaluation works

## Auto-graded flow

- MCQ and true/false immediately compare করা যায়
- Score এবং percentage সাথে সাথে calculate হয়
- Pass/fail check করা হয়
- Result save হয়
- Leaderboard update হয়

## Manual review flow

- Short answer or coding answer initially pending-review থাকে
- Teacher review page থেকে marks আর comments দেয়
- তারপর final score update হয়
- Then progress and ranking finalize করা হয়

## Progress + points system

“আমি শুধু raw score store করি নাই। Student-এর overall growth দেখানোর জন্য average score, total points, streak, passed/failed count, category-wise progress এবং global leaderboard maintain করেছি। Hard quiz হলে points multiplier বেশি হয়, তাই difficulty অনুযায়ী reward system আছে।”

## Security and control points

| Area            | What I used                               | Why                                      |
|-----------------|-------------------------------------------|------------------------------------------|
| Password        | bcryptjs                                  | Plain text password avoid করার জন্য      |
| Authentication  | passport-local + Google OAuth             | Secure login flow                        |
| Authorization   | ensureAdmin, ensureTeacher, ensureStudent | Role based access control                |
| Blocked account | enforceActiveAccount                      | Blocked user যেন system use করতে না পারে |
| Sessions        | connect-mongo                             | Session persistent রাখা                  |
| Method override | _method                                   | EJS form থেকে PUT/PATCH/DELETE support   |

## Special improvement points I can mention

## Multi-tab session support

একই browser-এ আলাদা tab-এ আলাদা role/login manage করার

## Teacher-wise filter

Student specific teacher select করে শুধুই ওই teacher-এর exam

## Category grouping

Long mixed list না দেখিয়ে category section-এ exam show

custom middleware আছে।

দেখতে পারে।

করা হয়।

**Dark mode**

Shared theme toggle দিয়ে  
light/dark mode support আছে।

**Manual coding review**

Coding submission teacher  
manually review করে  
mark/comment দিতে পারে।

**In-memory dev fallback**

Local MongoDB না থাকলে  
development-এ memory server  
fallback আছে।

**Challenge vs solution**

| Challenge                                    | My solution                                           |
|----------------------------------------------|-------------------------------------------------------|
| Many teachers, many exams, student confusion | Teacher-wise filter + category grouping + type filter |
| All question types cannot be auto-check      | Short answer / coding kept as pending-review workflow |
| Need role-based safety                       | Middleware-based auth and authorization               |
| Same browser multiple account handling       | Custom tab session middleware                         |
| Admin control over bad users                 | Block/unblock feature + active account enforcement    |

## VIVA TALK TRACK

**Ready-made speaking script**

"Sir, project-এর main goal ছিল একটা practical online assessment platform বানানো যেখানে admin, teacher, আর student তিনটা role clearly separated থাকে। Admin teacher approval আর user control করে, teacher quiz তৈরি ও review করে, আর student exam দেয়, result দেখে, progress track করে। আমি শুধু simple quiz না, manual review লাগতে পারে এমন short-answer আর coding submission flow-ও add করেছি। এছাড়া student confusion কমানোর জন্য teacher-wise filter, category-wise grouping, dark mode, multi-tab session, block/unblock control এই extra usability and control features add করেছি।"

**Possible viva questions with short answers****1. কেন MongoDB use করেছ?**

কারণ quiz, question, attempt, submission এগুলো document-style data। Nested structure easily store করা যায়, আর Mongoose দিয়ে schema + relation manage easy হয়েছে।

**2. কেন EJS use করেছ, React না?**

এই project-এর জন্য server-rendered flow enough ছিল। EJS দিয়ে দ্রুত page rendering, form handling, and MVC integration সহজ হয়েছে।

**3. Auto grading আর manual review দুটো কেন লাগল?**

সব answer machine evaluate করতে পারে না। MCQ/true-false auto-check হয়, কিন্তু short answer আর coding submission teacher judgement দরকার।

**4. Teacher-wise filter কেন important?**

একই platform-এ অনেক teacher exam দিলে student confuse হতে পারে। Teacher filter দিলে student intended teacher-এর exam সহজে খুঁজে পায়।

**5. Security কীভাবে handle করেছ?**

Password hash with bcrypt, Passport auth, role middleware, blocked-account enforcement, MongoDB session store, and route-level protection use করেছি।

**6. Project-এর unique point কী?**

শুধু quiz platform না, এখানে role approval, manual review, coding module, leaderboard, progress, multi-tab login handling, and teacher/category based browsing একসাথে আছে।

## If asked “future improvement?”

### Good future ideas

- Real coding execution sandbox
- Email notification for review/result
- More detailed analytics charts
- Question bank reuse
- CSV/PDF report export

### Nice final line

“Overall, project টার focus ছিল practical exam managementকে clean, role-based, and easy to use করা। আমি শুধু CRUD না, full workflow চিন্তা করে build করেছি — from teacher approval to student attempt, review, progress, and ranking.”

## Very short revision checklist

[Goal](#)[Tech stack](#)[3 roles](#)[Quiz flow](#)[Manual review](#)[Teacher filter](#)[Security](#)[Future work](#)